Mental Health Portal

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Chapter 1

Introduction

1.1 Background

Mental health has become one of the most important healthcare concerns worldwide. Traditional mental healthcare systems often rely on paper records, manual appointment scheduling, and physical documentation, making it difficult for therapists and patients to efficiently manage treatment records. Digital healthcare platforms provide an opportunity to improve accessibility, organization, and communication between therapists and patients.

The Mental Health Portal developed in this project provides a secure web-based platform where patients can schedule therapy sessions, record mood logs, receive follow-up reminders, and communicate through structured therapy sessions. Therapists can manage appointments, review patient mood trends, maintain encrypted session notes, and schedule follow-up sessions. Administrators supervise therapists, verify professional accounts, and manage therapy specializations.

The project demonstrates the practical implementation of relational database concepts using Laravel and MySQL while following proper database normalization and CRUD operations.

1.2 Problem Statement

Traditional methods of managing mental healthcare involve handwritten records, manual appointment booking, and unorganized patient histories. These approaches increase the chances of scheduling conflicts, misplaced records, delayed follow-up sessions, and inefficient communication between therapists and patients.

The objective of this project is to replace these manual processes with a centralized web application capable of securely storing patient information, managing appointments, tracking mood logs, maintaining session records, and simplifying administrative tasks.

1.3 Objectives

The main objectives of this project are:

- Develop a secure Mental Health Portal using Laravel Framework.

- Design a normalized relational database.
- Implement complete CRUD operations.
- Allow patients to register and manage therapy sessions. • Allow therapists to manage patients and therapy notes.
- Allow administrators to verify therapists and manage specializations.
- Demonstrate authentication using multiple user roles.
- Implement secure password hashing using Laravel Authentication.
- Develop an efficient appointment booking system.
- Provide mood tracking and follow-up management.

1.4 Scope

The project provides three different user roles.

1. Patient
    - Register
    - Login
    - Book Appointment
    - Mood Tracking
    - View Follow-ups
    - Edit Profile
2. Therapist
    - Login
    - View Assigned Patients
    - Create Session Notes
    - View Mood Logs

- Reschedule Sessions
- Create Follow-ups
- Cancel Sessions

3. Administrator

- Verify Therapists
- Manage Specializations
- View Patients
- View Therapists
- Monitor Appointments

1.5 Tools and Technologies

Technology	Purpose
Laravel 12	Backend Framework
PHP 8.2	Server-side Programming
MySQL	Database
Bootstrap 5	Frontend Design
Blade Templates	User Interface
HTML5	Structure
CSS3	Styling
JavaScript	Client-side Validation
Git	Version Control
GitHub	Source Code Repository

Table 1.1: Technologies Used

Chapter 2

System Analysis

2.1 Existing System

Traditional mental healthcare systems depend on manual appointment books and paper records. Patient information is often maintained in physical files, making it difficult to retrieve previous treatment history. Appointment scheduling is handled manually, resulting in scheduling conflicts and inefficient communication.

2.2 Proposed System

The proposed Mental Health Portal provides an integrated online platform for patients, therapists, and administrators.

The proposed system provides:

- Secure Authentication
- Appointment Booking
- Mood Tracking
- Follow-up Scheduling
- Therapist Verification
- Session Notes
- Patient Profile Management
- Dashboard for each user
- Complete CRUD Operations

2.3 Functional Requirements

Patient

- Register
- Login

- Book Appointment
- Create Mood Log
- Edit Mood Log
- Delete Mood Log
- Edit Profile
- View Follow-ups

Therapist

- Login
- View Assigned Patients
- View Patient Profile
- View Mood Trends
- Create Session Notes
- Cancel Appointment
- Reschedule Appointment
- Create Follow-up

Administrator

- Login
- Verify Therapist
- Manage Specializations
- View Patients

-
-

View Therapists

View Appointments

2.4 Non-functional Requirements

- Secure Authentication
- Fast Response Time
- Easy Navigation
- Responsive Design
- Data Integrity
- Scalability
- Maintainability
- Availability
- Reliability

Chapter 3

System Analysis and Design

3.1 System Requirements

The Mental Health Portal is designed as a secure web-based application to facilitate communication between patients, therapists, and administrators. The system enables appointment scheduling, mood tracking, secure clinical documentation, therapist verification, and follow-up management.

The system has three different user roles:

- Patient
- Therapist
- Administrator

Each role has different permissions and responsibilities.

3.2 Functional Requirements

The functional requirements of the system are listed below.

3.2.1 Patient Module

- Register a new patient account.
- Login securely.
- Update profile information.
- Book appointments with available therapists.
- View upcoming appointments.

View appointment history.

Cancel appointments.

-
-
- Record daily mood logs.
- Edit mood logs.
- Delete mood logs.
- View therapist follow-up requests.
- Confirm follow-up appointments.

3.2.2 Therapist Module

- Secure login.
- View assigned patients.
- View patient profile.
- View patient mood history.
- Create encrypted session notes.
- Schedule follow-ups.
- Cancel appointments.
- View upcoming appointments.

3.2.3 Administrator Module

- Login securely.
- Verify therapist accounts.
- Manage specializations.
- View all patients.
- View all therapists.
- View appointments.
- View cancellations.

- View follow-up records.

3.3 Non-Functional Requirements

The system also satisfies several non-functional requirements.

3.3.1 Security

- Passwords are encrypted using Laravel Hashing.
- Role-based authentication is implemented.
- Unauthorized users cannot access protected routes.
- Session notes remain encrypted.

3.3.2 Performance

- Efficient SQL queries are used.
- Eloquent Relationships reduce redundant database calls.
- Frequently accessed records are loaded using eager loading.

3.3.3 Reliability

- Database transactions prevent partial updates.
- Validation prevents invalid data insertion.
- Foreign key constraints maintain data integrity.

3.3.4 Usability

- Responsive Bootstrap interface.
- Easy navigation for all users.
- Simple appointment booking process.

-
-

3.4 Conceptual Schema

The database consists of the following major entities:

- Patient
- Therapist

Administrator

Appointment

- SessionNote
- MoodLog
- FollowUp
- Cancellation
- Specialization

The relationships between these entities are summarized below.

Table 3.1: Entity Relationships

Relationship	Cardinality
Patient – Appointment	One-to-Many
Therapist – Appointment	One-to-Many
Appointment – SessionNote	One-to-Many
Appointment – FollowUp	One-to-Many
Appointment – Cancellation	One-to-One
Patient – MoodLog	One-to-Many
Specialization – Therapist	One-to-Many

Figure 2.1 shows the finalized ER Diagram of the system.

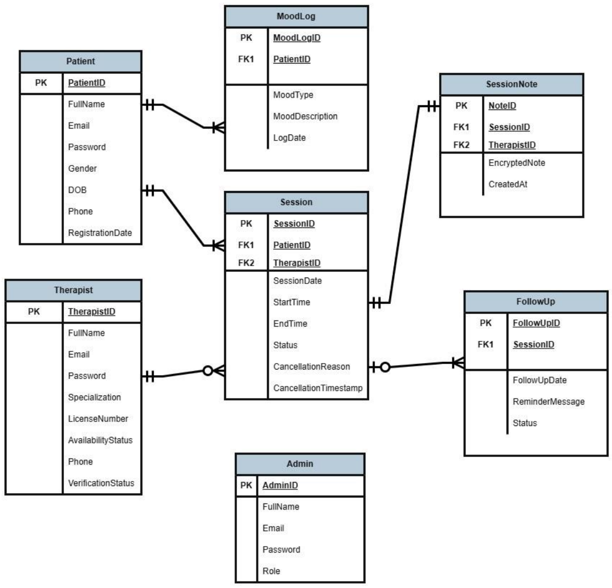

Figure 3.1: Finalized ER Diagram

3.5 Normalized Relations

The database was normalized up to Third Normal Form (3NF) to eliminate redundancy and maintain consistency.

3.5.1 Patient

Patient(
PatientID PK,
FullName,
Email,
Password,
Gender,
DateOfBirth,
Phone,
Address
)

3.5.2 Therapist

Therapist(
TherapistID PK,
FullName,
Email,
Password,
Phone,
LicenseNumber,
AvailabilityStatus,
VerificationStatus, SpecializationID FK
)

3.5.3 Specialization

Specialization(
SpecializationID PK,
SpecializationName
)

3.5.4 Appointment

Appointment(
SessionID PK,
PatientID FK,
TherapistID FK,
SessionDate,
StartTime,
EndTime,
Status
)

3.5.5 SessionNote

SessionNote(
NoteID PK,
SessionID FK,
EncryptedNote,
CreatedAt
)

3.5.6 MoodLog

MoodLog(
MoodLogID PK,
PatientID FK,
MoodType,
MoodDescription,
LogDate
)

3.5.7 FollowUp

FollowUp(
FollowUpID PK,
SessionID FK, FollowUpDate,
ReminderMessage,
Status
)

3.5.8 Cancellation

Cancellation(
CancellationID PK,
SessionID FK,
CancellationReason,
CancellationTimestamp
)

3.5.9 Administrator

Admin(
AdminID PK,
FullName,
Email,
Password
)

The database satisfies First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF). All non-key attributes depend only on the primary key, and transitive dependencies have been removed to ensure data integrity and reduce redundancy.

Chapter 4

Database Design

4.1 Database Overview

The Mental Health Portal uses a MySQL relational database to securely store patient information, therapist records, appointments, mood logs, follow-ups, cancellations, and session notes. The database was designed according to Third Normal Form (3NF), ensuring minimal redundancy and maintaining referential integrity through primary and foreign key constraints.

The database consists of the following major tables:

- Admin
- Patient
- Therapist
- Specialization
- Appointment
- SessionNote
- MoodLog
- FollowUp
- Cancellation

4.2 SQL Database Tables

This section presents the final database schema implemented in MySQL.

4.2.1 Admin Table

```
CREATE TABLE Admin(
```

```
    AdminID INT PRIMARY KEY AUTO_INCREMENT,
    FullName VARCHAR(100),
    Email VARCHAR(100) UNIQUE,
    Password VARCHAR(255)
);
```

4.2.2 Patient Table

```
CREATE TABLE Patient(
    PatientID INT PRIMARY KEY AUTO_INCREMENT,
    FullName VARCHAR(100),
    Email VARCHAR(100) UNIQUE,
    Password VARCHAR(255),
    Gender VARCHAR(20),
    DateOfBirth DATE,
    Phone VARCHAR(20),
    Address TEXT
);
```

4.2.3 Specialization Table

```
CREATE TABLE Specialization(
    SpecializationID INT PRIMARY KEY AUTO_INCREMENT,
    SpecializationName VARCHAR(100)
);
```

4.2.4 Therapist Table

```
CREATE TABLE Therapist(
    TherapistID INT PRIMARY KEY AUTO_INCREMENT,
    FullName VARCHAR(100),
    Email VARCHAR(100) UNIQUE,
    Password VARCHAR(255),
    Phone VARCHAR(20),
    LicenseNumber VARCHAR(50),
    AvailabilityStatus VARCHAR(30),
    VerificationStatus VARCHAR(30),
    SpecializationID INT,

    FOREIGN KEY (SpecializationID)
```

```
    REFERENCES Specialization(SpecializationID)
);
```

4.2.5 Appointment Table

```
CREATE TABLE Appointment(
    SessionID INT PRIMARY KEY AUTO_INCREMENT,
    PatientID INT,
    TherapistID INT,
    SessionDate DATE,
    StartTime TIME,
    EndTime TIME,
    Status VARCHAR(30),

    FOREIGN KEY (PatientID)
    REFERENCES Patient(PatientID),

    FOREIGN KEY (TherapistID)
    REFERENCES Therapist(TherapistID)
);
```

4.2.6 SessionNote Table

```
CREATE TABLE SessionNote(
    NoteID INT PRIMARY KEY AUTO_INCREMENT,
    SessionID INT,
    EncryptedNote TEXT,
    CreatedAt TIMESTAMP,

    FOREIGN KEY (SessionID)
    REFERENCES Appointment(SessionID)
);
```

4.2.7 MoodLog Table

```
CREATE TABLE MoodLog(
    MoodLogID INT PRIMARY KEY AUTO_INCREMENT, PatientID INT,
    MoodType VARCHAR(50),
    MoodDescription TEXT,
    LogDate DATE,
```

```
    FOREIGN KEY (PatientID)
    REFERENCES Patient(PatientID)
);
```

4.2.8 FollowUp Table

```
CREATE TABLE FollowUp(
    FollowUpID INT PRIMARY KEY AUTO_INCREMENT,
    SessionID INT,
    FollowUpDate DATE,
    ReminderMessage TEXT,
    Status VARCHAR(30),

    FOREIGN KEY (SessionID)
    REFERENCES Appointment(SessionID)
);
```

4.2.9 Cancellation Table

```
CREATE TABLE Cancellation(
    CancellationID INT PRIMARY KEY AUTO_INCREMENT,
    SessionID INT,
    CancellationReason TEXT,
    CancellationTimestamp TIMESTAMP,

    FOREIGN KEY (SessionID)
    REFERENCES Appointment(SessionID)
);
```

4.3 SQL Queries Used

The project makes extensive use of SQL queries through Laravel Eloquent ORM. Some representative queries are shown below.

4.3.1 Insert Query

Patient appointment booking:

```
INSERT INTO Appointment
(PatientID, TherapistID, SessionDate,
```

```
StartTime, EndTime, Status)

VALUES
(1,2,'2026-07-10','10:00','11:00','Scheduled');
```

4.3.2 Update Query

Therapist verification:

```
UPDATE Therapist

SET VerificationStatus='Verified'

WHERE TherapistID=3;
```

4.3.3 Delete Query

Delete a mood log:

```
DELETE FROM MoodLog

WHERE MoodLogID=5;
```

4.3.4 Simple Select Query

Retrieve all therapists:

```
SELECT *

FROM Therapist;
```

4.3.5 Select with Condition

Retrieve appointments of one patient:

```
SELECT *

FROM Appointment

WHERE PatientID=1;
```

4.3.6 Join Query

Retrieve appointment details with therapist and patient names:

```
SELECT

Appointment.SessionID,

Patient.FullName,

Therapist.FullName,

Appointment.SessionDate

FROM Appointment

JOIN Patient
ON Appointment.PatientID = Patient.PatientID

JOIN Therapist
ON Appointment.TherapistID = Therapist.TherapistID;
```

4.3.7 Aggregate Query

Count appointments of each therapist:

```
SELECT

TherapistID,

COUNT(*) AS TotalAppointments

FROM Appointment

GROUP BY TherapistID;
```

4.3.8 Mood Trend Query

Count each mood type:

```
SELECT

MoodType,
```

```
COUNT(*) AS Total

FROM MoodLog

GROUP BY MoodType;
```

4.3.9 Follow-up Query

Retrieve pending follow-ups:

```
SELECT *

FROM FollowUp

WHERE Status='Pending';
```

4.4 Database Features

The implemented database provides the following features:

- Primary keys uniquely identify every record.
- Foreign keys maintain referential integrity.
- Duplicate patient mood logs for the same date are prevented.
- Therapist appointment conflicts are checked before booking.
- Session notes are securely stored.
- Therapist verification is controlled by the administrator.
- Follow-ups are linked directly with appointments.
- Appointment cancellation history is permanently maintained.

The database structure ensures consistency, scalability, and efficient retrieval of information while supporting all system modules developed using Laravel.

Chapter 5

System Implementation

5.1 Introduction

The Mental Health Portal was developed using the Laravel PHP Framework following the MVC (Model-View-Controller) architecture. Bootstrap 5 was used to design a responsive user interface, while MySQL served as the backend database. Laravel Eloquent ORM was used to interact with the database, providing efficient CRUD operations for all modules.

The application consists of three major user roles:

- Patient
- Therapist
- Administrator

Each module is discussed below.

5.2 Authentication Module

The authentication system allows users to securely register and log into the application. Passwords are encrypted using Laravel Hashing (Bcrypt), and users are redirected to dashboards according to their roles.

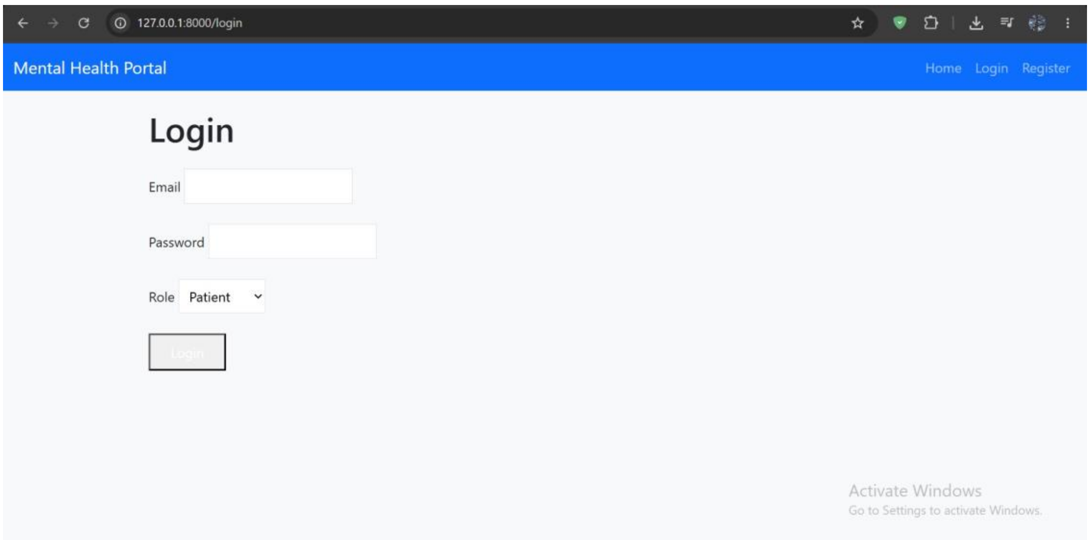

Figure 5.1: Login Page

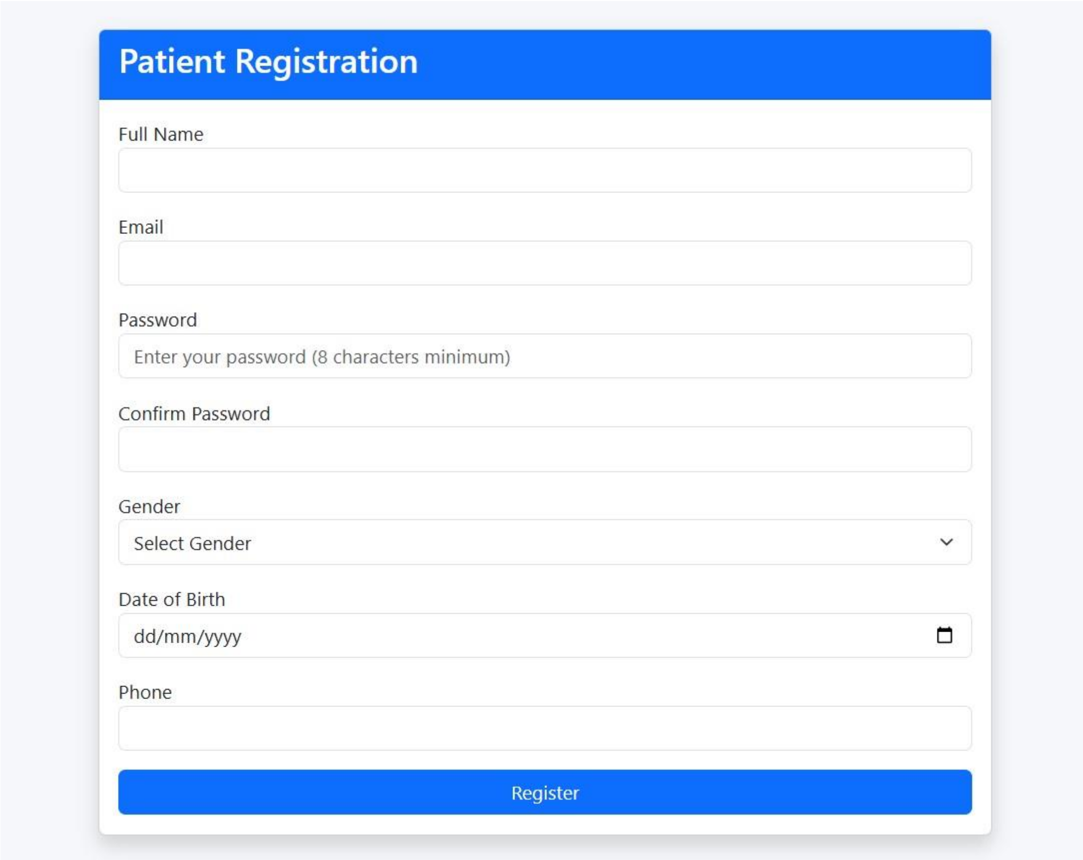

Figure 5.2: Registration Page

5.3 Patient Dashboard

After successful login, the patient is redirected to the Patient Dashboard. The dashboard allows the patient to:

- View upcoming appointments.
- View previous appointments.
- Record mood logs.
- Edit mood logs.
- Delete mood logs.
- Book appointments.
- View therapist follow-ups.
- Update personal profile.

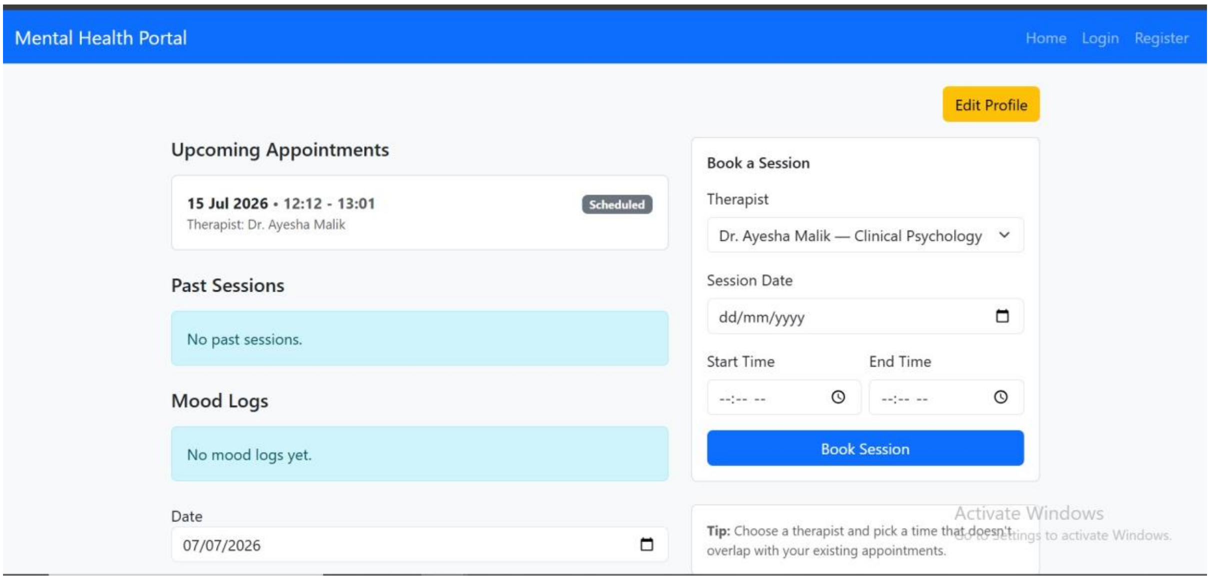

Figure 5.3: Patient Dashboard

5.4 Appointment Booking (Create)

Patients can schedule appointments by selecting an available therapist, choosing a date, and specifying start and end times.

The system validates:

- Therapist availability.
- Date cannot be in the past.
- End time must be greater than start time.
- No overlapping appointments.

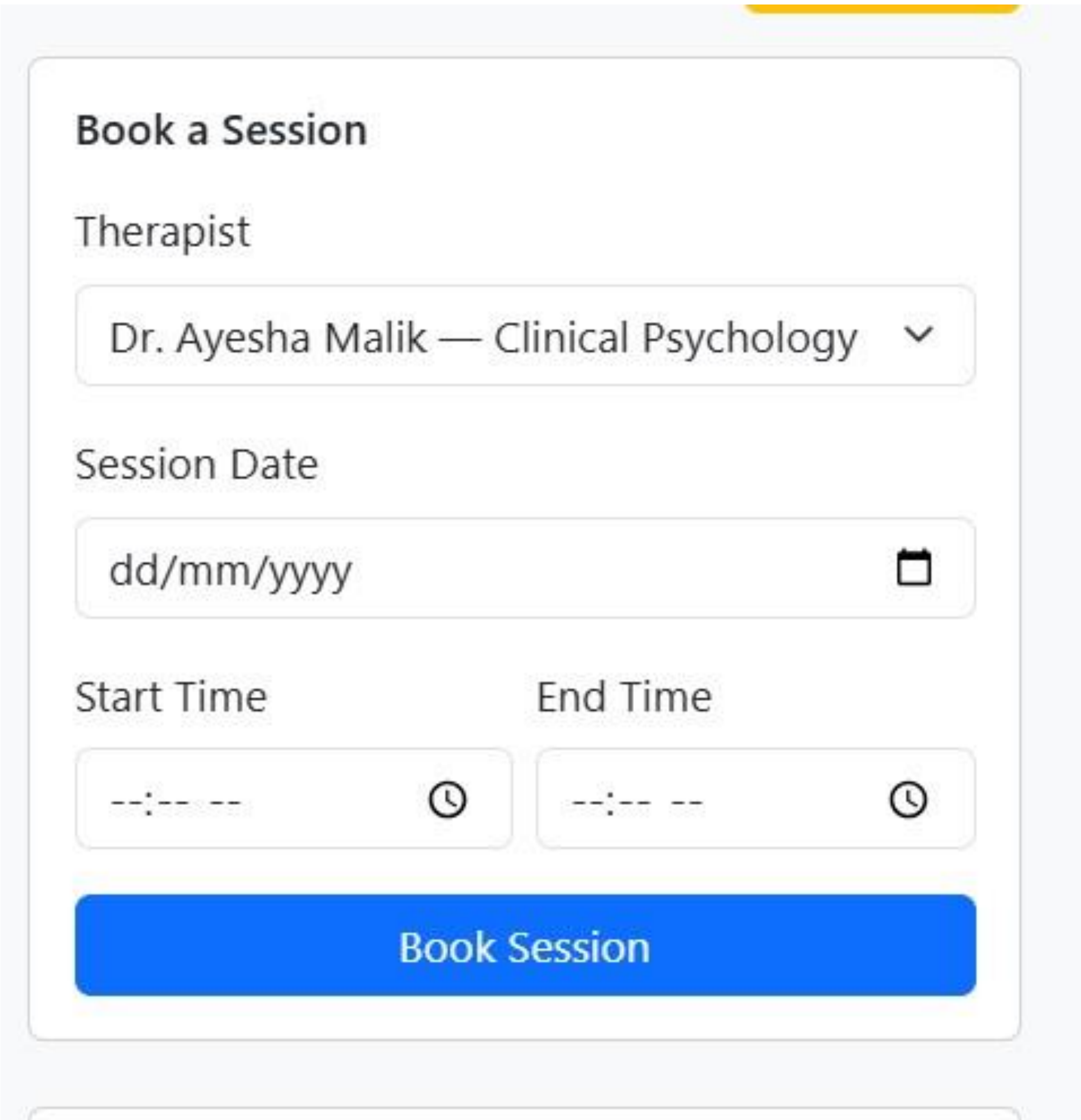

Figure 5.4: Book Appointment

5.5 Mood Log Module (CRUD)

Patients can maintain a personal mood diary.
The CRUD operations implemented include:

- Create Mood Log
- Read Mood Logs
- Update Mood Log
- Delete Mood Log

Duplicate mood entries for the same date are prevented.

5.5.1 Read Mood Logs

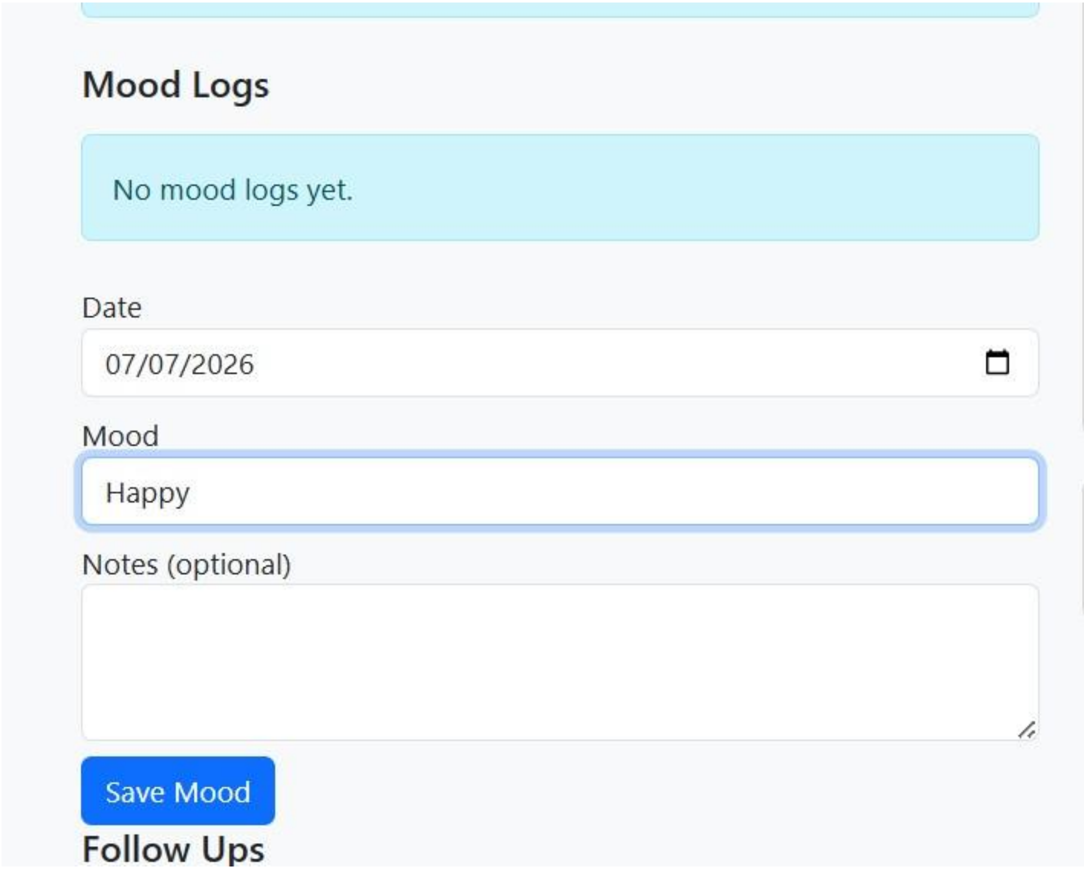

Figure 5.5: View Mood Logs

5.6 Patient Profile

Patients can update their personal information at any time.
Editable information includes:

- Full Name
- Email
- Phone Number
- Address
- Password

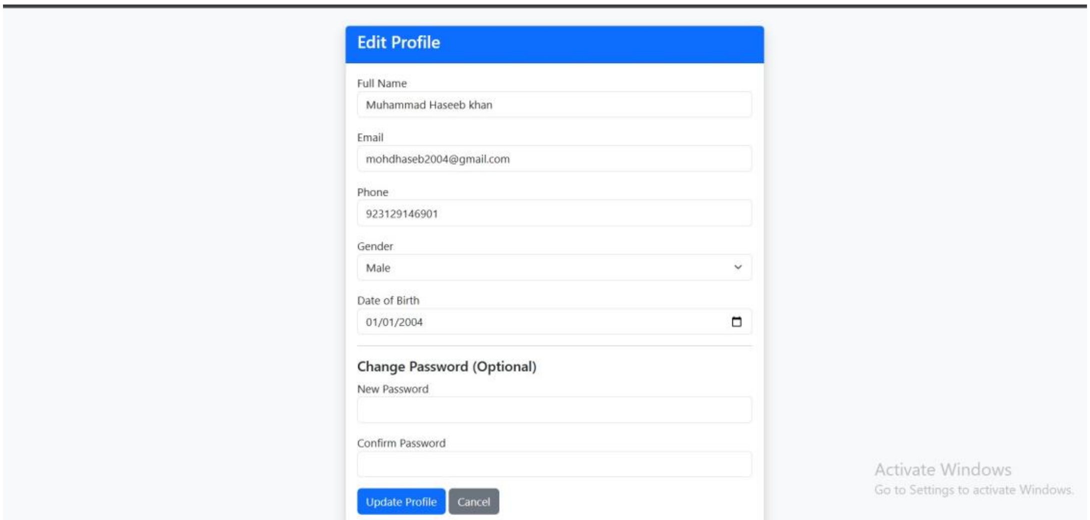

Figure 5.6: Edit Patient Profile

5.7 Therapist Dashboard

After login, therapists can monitor appointments and patients.

The therapist dashboard provides:

- Upcoming appointments
- Assigned patients
- Session note creation
- Appointment cancellation
- Follow-up creation
- Mood log analysis

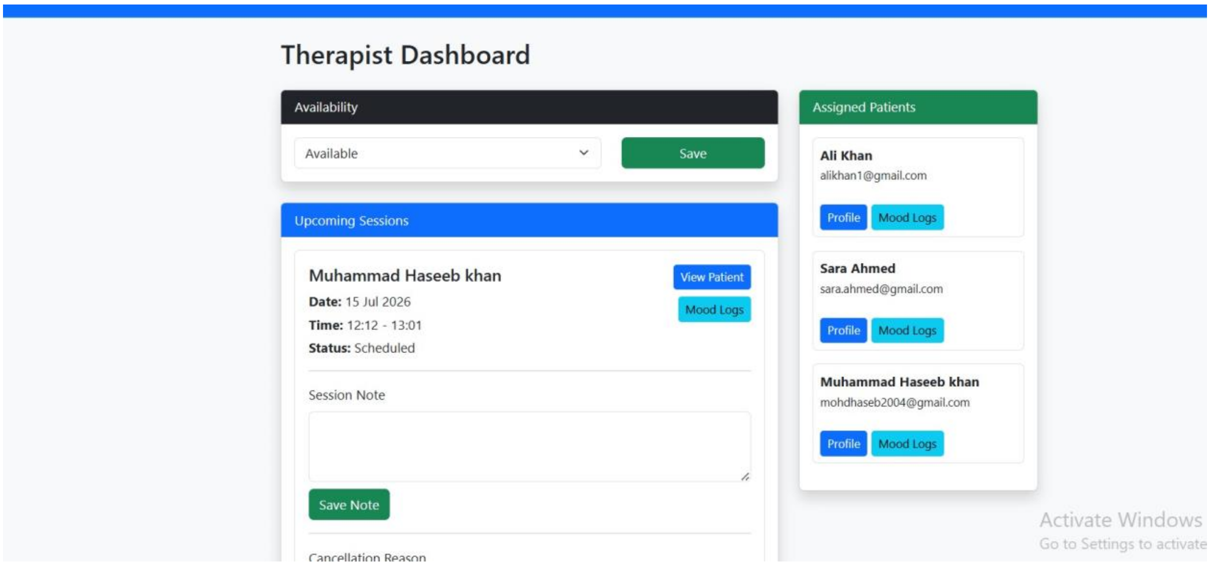

Figure 5.7: Therapist Dashboard

5.8 Patient Profile View

Therapists can access patient history.

The page displays:

- Patient information
- Appointment history
- Mood history
- Session notes
- Follow-ups

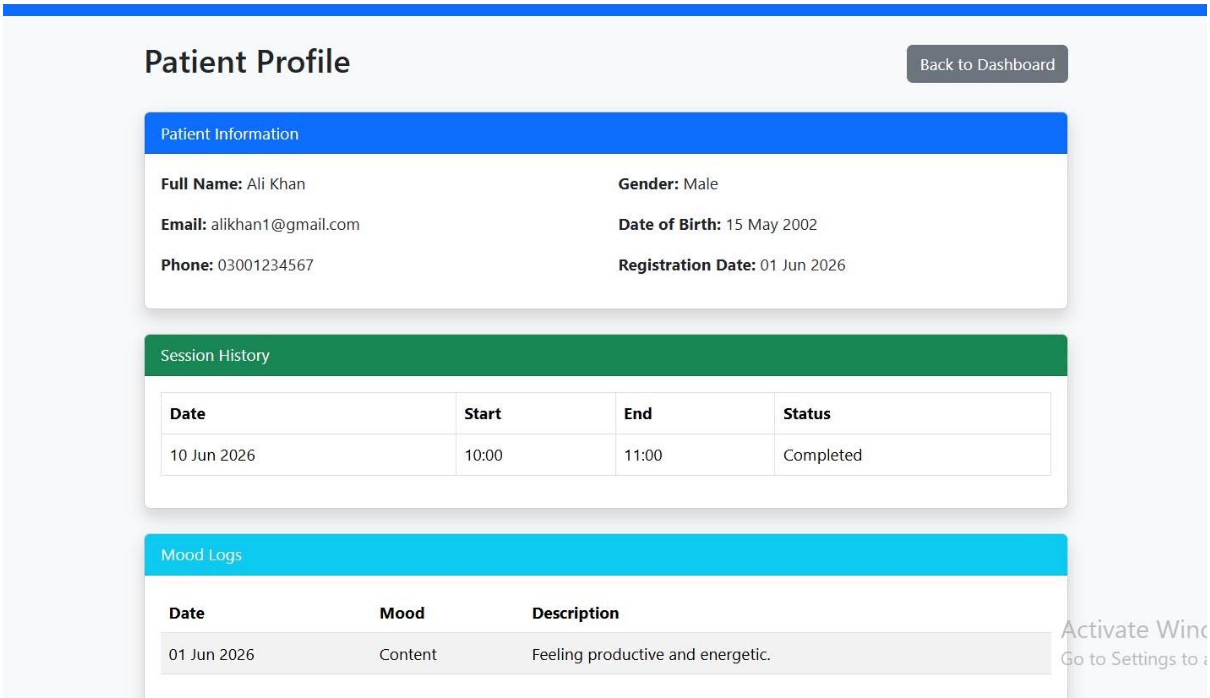

Figure 5.8: Therapist Viewing Patient Profile

5.9 Session Notes

Therapists can securely save encrypted notes after each session.
These notes are only accessible by authorized therapists.

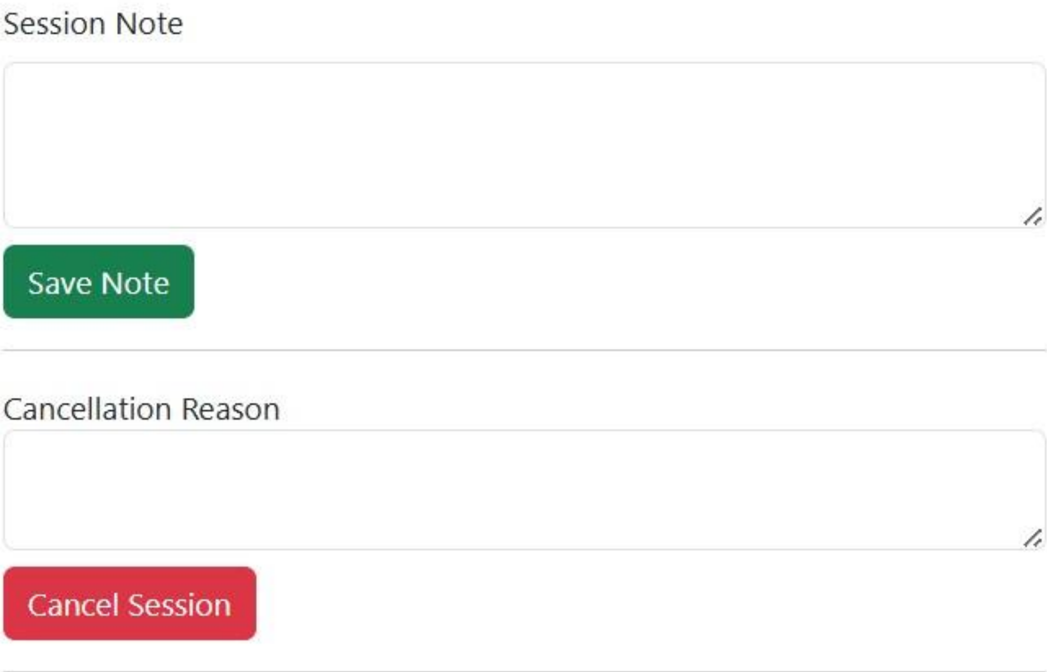

Figure 5.9: Encrypted Session Notes

5.10 Follow-up Module

Therapists can create follow-up sessions after appointments.

Patients can later confirm these follow-ups, automatically generating a new appointment.

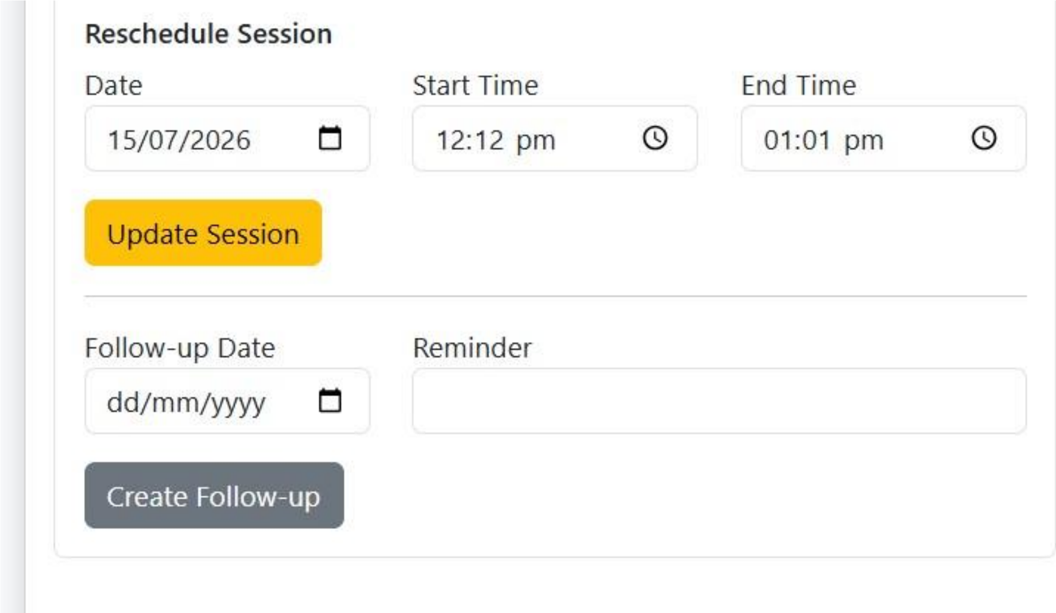

Figure 5.10: Follow-up Creation

5.11 Mood Trend Analysis

Therapists can view graphical summaries of patient mood trends.

This assists therapists in evaluating patient progress over time.

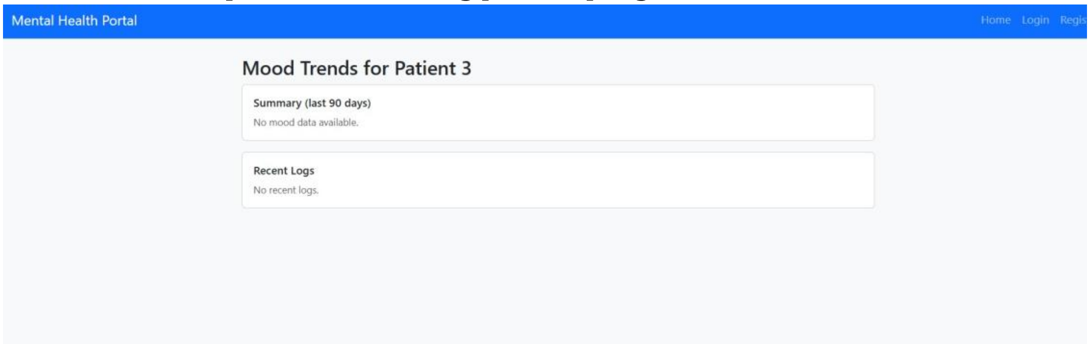

Figure 5.11: Mood Trend Analysis

5.12 Administrator Dashboard

The administrator has complete control over the system.

Functions include:

- View all patients
- View therapists
- Verify therapist accounts
- Manage specializations
- View appointments
- View follow-ups
- View cancellations

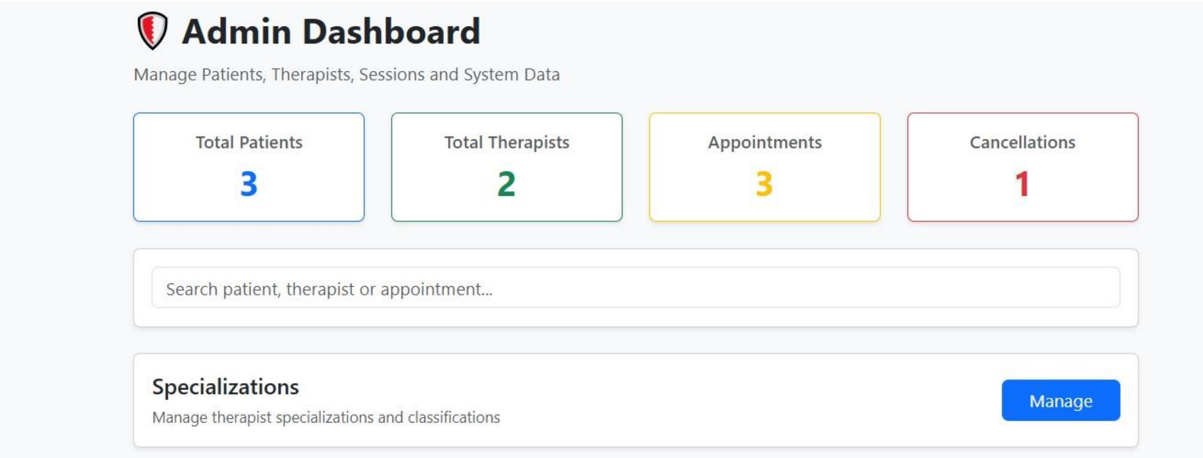

Figure 5.12: Administrator Dashboard

5.13 Therapist Verification

Administrators verify therapist accounts before allowing them to receive appointments.

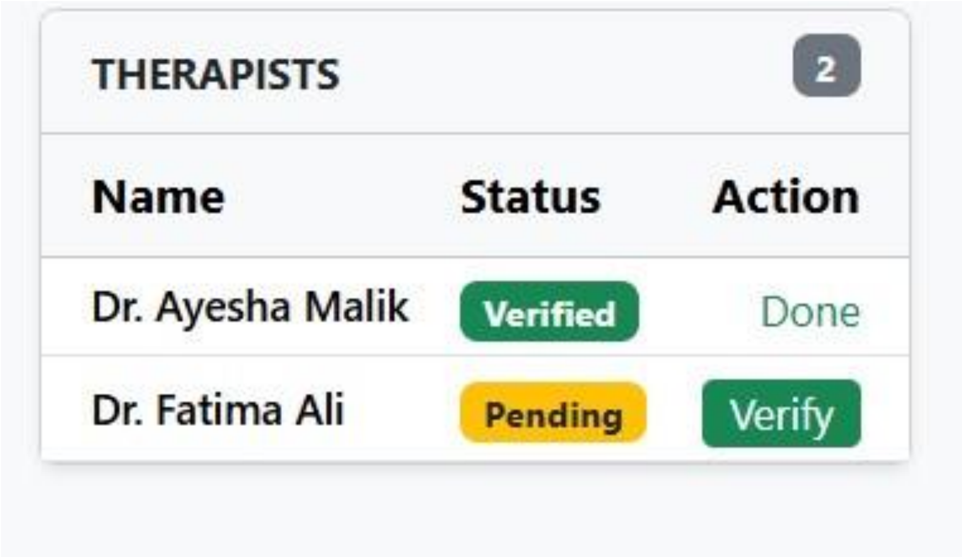

Figure 5.13: Therapist Verification

5.14 Specialization Management (CRUD)

Administrators can manage therapist specializations.
CRUD operations include:

- Create Specialization
- View Specializations
- Update Specialization
- Delete Specialization

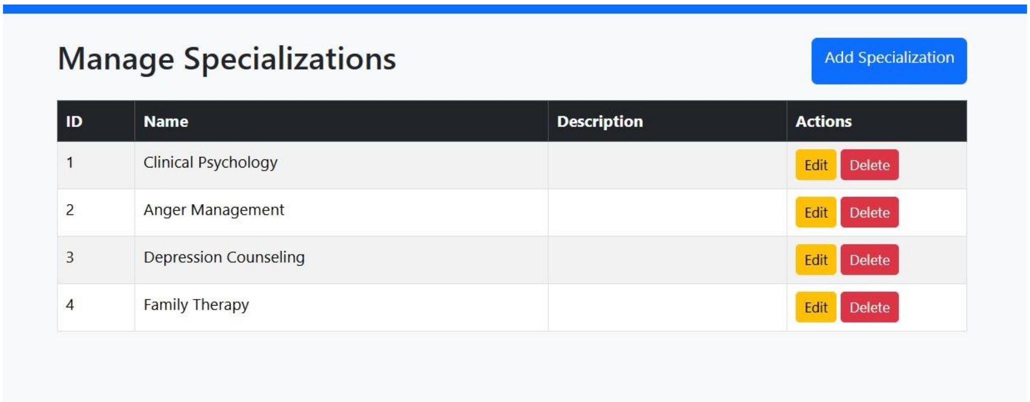

Figure 5.14: Specialization Management

5.15 Summary

The system successfully implements all required CRUD operations using Laravel controllers, models, Blade templates, and MySQL. Input validation, authentication, authorization, and database integrity constraints ensure secure and reliable operation. Bootstrap provides a responsive interface, while Laravel Eloquent simplifies interaction with the database.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

The Mental Health Portal was successfully designed and implemented using the Laravel Framework and MySQL database management system. The project fulfills the objectives of providing a secure, user-friendly, and efficient platform for communication between patients, therapists, and administrators.

The system supports complete CRUD (Create, Read, Update, Delete) operations across all major modules, including appointment management, mood logging, follow-up scheduling, therapist verification, and specialization management. Laravel's MVC architecture, Eloquent ORM, and Blade templating engine were used to ensure modularity, maintainability, and scalability.

Authentication and authorization mechanisms provide secure access based on user roles, ensuring that patients, therapists, and administrators can only access functionalities relevant to them. Passwords are securely encrypted using Laravel's hashing mechanism, while server-side validation maintains data integrity and protects against invalid user input.

The database was carefully normalized to Third Normal Form (3NF), eliminating redundancy and ensuring efficient storage and retrieval of information. Relationships between entities were implemented using foreign key constraints, maintaining consistency throughout the application.

Overall, the developed system demonstrates the practical application of database concepts, web development technologies, and software engineering principles. It provides a solid foundation for a real-world mental healthcare management platform.

6.2 Achievements

The following objectives were successfully achieved during the development of the project:

- Designed a normalized relational database.
- Implemented secure user authentication and role-based authorization.
- Developed separate dashboards for patients, therapists, and administrators.

- Implemented appointment booking with overlap detection.
- Developed a complete Mood Log management system.
- Implemented encrypted session notes.
- Developed therapist follow-up management.
- Implemented therapist verification by administrators.
- Developed CRUD functionality for specialization management.
- Integrated Laravel Eloquent ORM with MySQL database.
- Designed a responsive Bootstrap user interface.

6.3 Limitations

Although the developed system satisfies the project requirements, a few limitations remain:

- No online payment gateway is integrated.
- Email notifications are not implemented.
- Video consultation is not supported.
- No mobile application is available.
- Mood trend analysis is limited to basic statistics.

6.4 Future Work

Several enhancements can be incorporated into future versions of the system to improve its functionality and user experience.

- Integration of secure video consultation using WebRTC or Zoom APIs.
- Online payment integration through Stripe or PayPal.
- Email and SMS reminders for appointments and follow-ups.

- AI-powered mood prediction and mental health recommendations.
- Advanced analytics dashboards with graphical reports.
- Real-time chat between therapists and patients.
- Mobile applications for Android and iOS.
- Multi-language support.
- File upload functionality for prescriptions and medical reports.
- Cloud deployment using AWS, Azure, or DigitalOcean.

6.5 Final Remarks

The Mental Health Portal demonstrates how modern web technologies and database management systems can be combined to build a secure and efficient healthcare application. The project not only satisfies the academic requirements of the DBMS laboratory but also represents a practical solution that can be further extended into a production-ready mental healthcare platform.